Stochastic Processes in Finance

An Empirical Comparison of AAPL and VIX

Abstract

This study compares a geometric random walk, geometric Brownian motion, and an exponentiated Ornstein–Uhlenbeck process with observed AAPL stock prices and the VIX volatility index. Each asset is studied using 504 daily log returns and 200 diagnostic paths per model, plus a separate 10,000-path Monte Carlo experiment for uncertainty, event probabilities and convergence. The comparison considers return moments, serial dependence, stationarity diagnostics, and drawdown. The exercise evaluates in-sample descriptive fit; it does not establish predictive performance or market efficiency.

1 Research objective and hypotheses

The objective is to assess which empirical features of two contrasting financial series can be reproduced by simple stochastic processes. VIX is a volatility index, not a directly investable stock price; applying price-process models to it is an illustrative statistical exercise.

The original research questions are expressed as four hypotheses:

1. **H1: Weak return dependence.** Returns have little serial correlation. Autocorrelation is examined as a limited diagnostic of the broader weak-form efficiency question.
2. **H2: Random-walk approximation.** A geometric random walk reproduces observed return and risk characteristics reasonably well.
3. **H3: GBM distributional fit.** Gaussian log-return increments provide an adequate description of the observed distribution.
4. **H4: Mean reversion.** Log-price stationarity is consistent with using a mean-reverting process. An ADF test is a diagnostic, not proof of OU dynamics.

2 Data and methodology

The analysis uses the local snapshots `Data/Assets/stock.csv` and `Data/Assets/index.csv`, originally sourced through yfinance. Each series contributes its last 505 closing observations, giving 504 log returns, $r_t = \log(S_t/S_{t-1})$. The CSV files contain no date column, so exact calendar coverage cannot be reconstructed from these inputs. The report is rebuilt from both snapshots without downloading fresh data.

Each asset uses 200 paths of 504 increments, $\Delta t = 1/252$, and random seed 51 reset separately for reproducibility. Inputs are estimated from the same sample used for comparison, making this an in-sample study. Annualized mean and volatility use $m = 252\bar{r}$ and $\sigma = \sqrt{252}\,\mathrm{std}(r)$, with population standard deviation.

2.1 Implemented processes

Let Z_t be independent standard normal draws and $X_t = \log S_t$:

$$\begin{aligned} \text{RW:} \quad & S_{t+1} = S_t \exp(\sqrt{\Delta t} Z_t), \\ \text{GBM:} \quad & S_{t+1} = S_t \exp[m\Delta t + \sigma\sqrt{\Delta t} Z_t], \\ \text{Log-OU:} \quad & X_{t+1} = X_t + \theta(\ell - X_t)\Delta t + \sigma\sqrt{\Delta t} Z_t, \quad S_t = e^{X_t}. \end{aligned}$$

Here ℓ is the sample mean log price and θ is specified rather than fitted. The random walk fixes annual log volatility at one. The GBM simulator expects arithmetic drift μ. We now supply $\mu = m + \sigma^2/2$, so its update has annual log drift m. This corrects the previous convention, which applied an extra variance subtraction to the estimated log-return drift. This is an in-sample calibration, not an estimate of risk-neutral drift.

Table 1: Model inputs. Starting levels retain their original units.

Asset	S_0	m	σ	ℓ	θ
AAPL	168.05	0.2191	0.2814	5.4276	2.5
VIX	16.03	0.0910	1.4147	2.8723	3.0

2.2 Measurement conventions

Simulated return moments use pooled paths. Return ADF tests and correlations use flattened paths, creating artificial joins between independent simulations. Simulated price ADF p-values and drawdowns are averages of per-path statistics; average p-values are not combined tests. Observed and OU price ADF tests use log prices, whereas RW and GBM use levels. Pearson kurtosis equals three under normality. Drawdowns are negative losses relative to preceding peaks.

3 Results: AAPL

Table 2: AAPL: observed and simulated statistics.

Metric	AAPL	RW	GBM	Log-OU
Annualized mean log return	21.91%	0.47%	24.09%	15.04%
Annualized volatility	28.14%	99.74%	28.20%	28.23%
Return autocorrelation (lag 1)	0.05196	-0.0003034	-0.00425	-0.003942
ADF p-value: returns	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$
ADF p-value: price series	0.1798	0.4088	0.615	0.2197
Return skewness	0.5675	-0.001577	-0.005083	0.004781
Return kurtosis (Pearson)	13.77	2.994	2.973	2.998
Maximum drawdown	-33.36%	-77.41%	-26.75%	-28.68%

The sample contains 504 observed daily log returns. Its annualized mean log return is 21.91% and annualized volatility is 28.14%.

3.1 Distribution and risk

Observed Pearson kurtosis is 13.77, versus 2.97–3.00 across the simulated models. Observed skewness is 0.567. Gaussian shocks do not reproduce the observed higher return moments in this run. Observed maximum drawdown is -33.36%; model drawdowns are path averages and should not be treated as forecasts.

3.2 Persistence and model fit

Lag-one return autocorrelation is 0.052. This is a descriptive measure, not a formal test of market efficiency. The ADF test on observed log prices does not reject the unit-root null at the 5% level. That result alone does not identify an OU process. GBM simulated volatility is 28.20%, while its annualized mean log return is 24.09%. The log drift is calibrated to the sample mean; finite simulation samples still introduce Monte Carlo error.

RW: geometric random walk; GBM: geometric Brownian motion; Log-OU: exponentiated OU process.

AAPL: Price dynamics

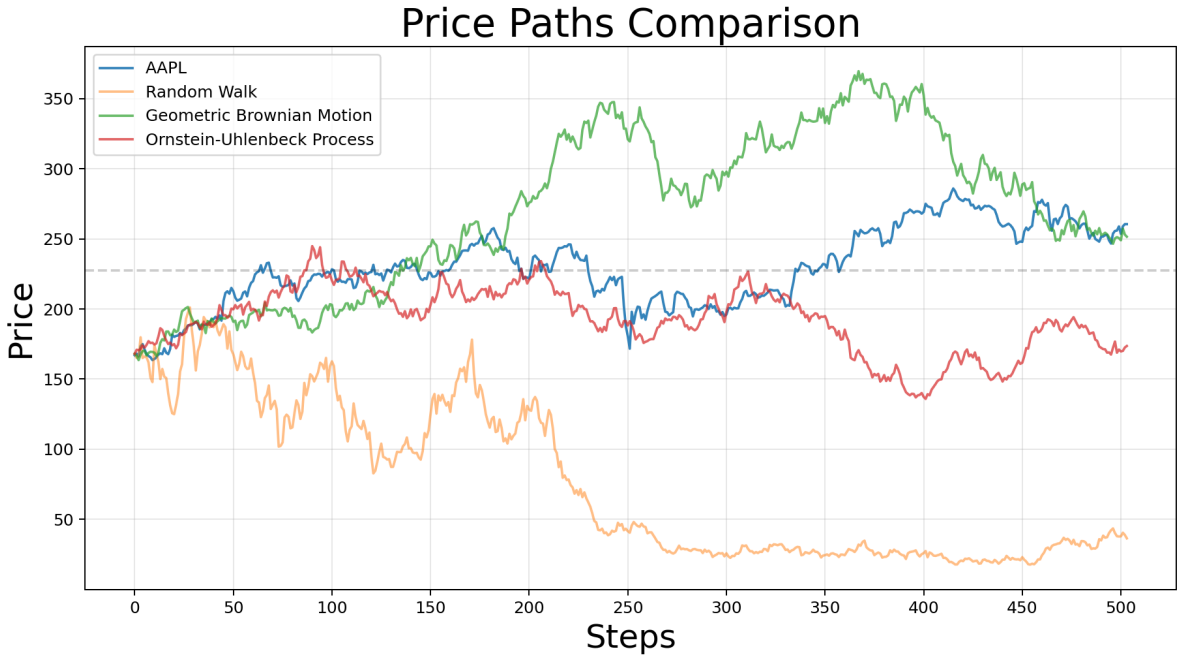


Figure 1: AAPL: Observed series and one illustrative path per model.

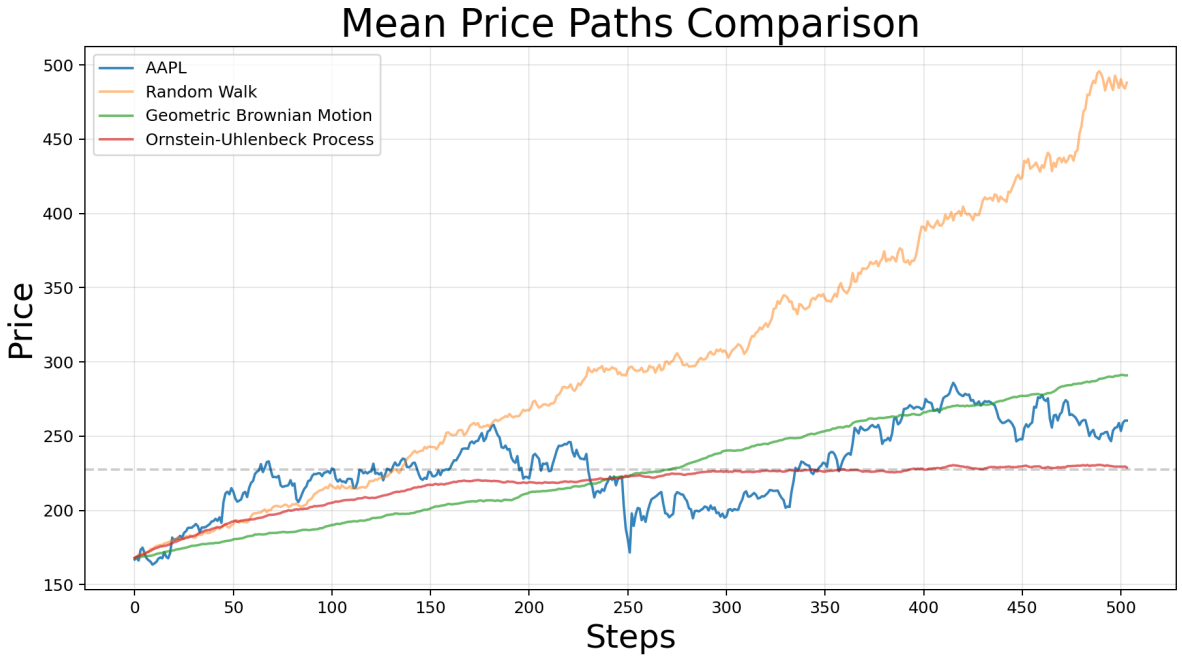


Figure 2: AAPL: Observed series and ensemble mean of 200 simulated paths.

AAPL: Return diagnostics

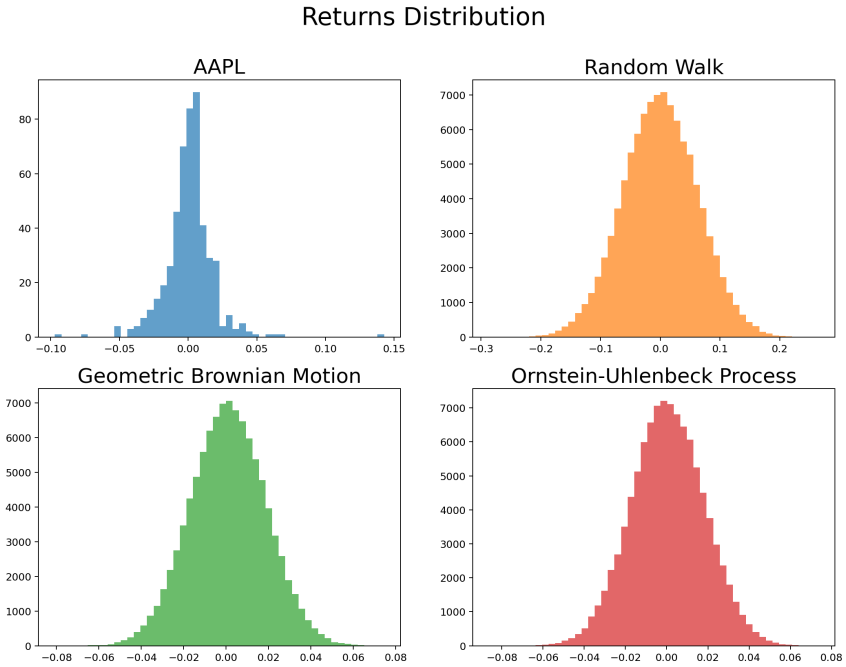


Figure 3: AAPL: Observed and simulated log-return distributions; counts use different sample sizes.

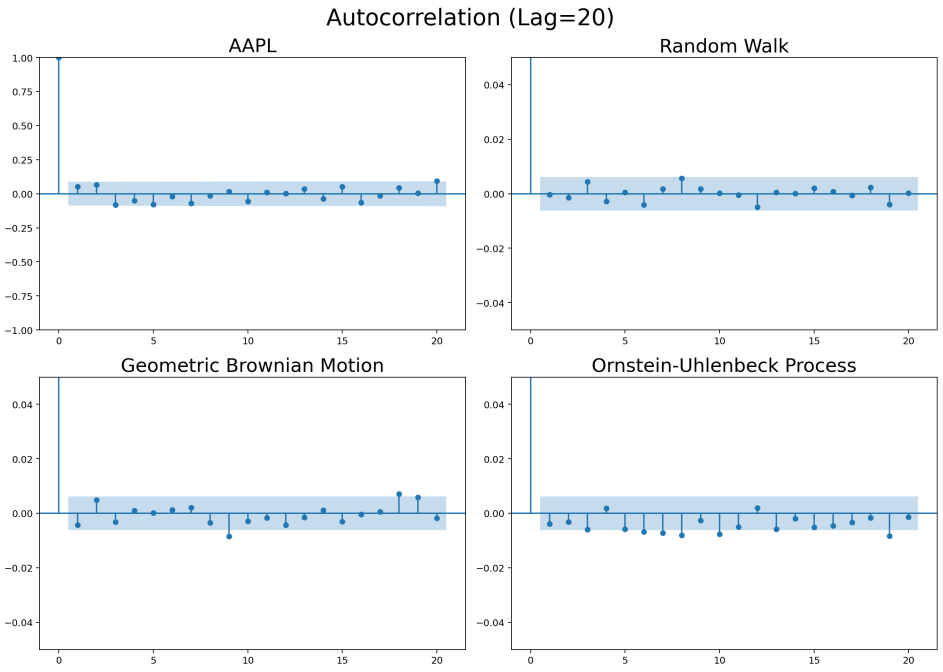


Figure 4: AAPL: Log-return autocorrelations; simulated diagnostics pool independent paths.

AAPL: Monte Carlo uncertainty and risk

A separate sample of 10,000 paths per model uses seed 52 and a 504-step horizon ($T = 2$ trading years). Parameters are held fixed. These are conditional model scenarios, not validated forecasts. VIX changes describe index levels, not investable returns.

For each path, terminal proportional change and worst drawdown are

$$R_T^{(i)} = S_T^{(i)}/S_0 - 1, \qquad D^{(i)} = \min_t \left(S_t^{(i)} / \max_{u \le t} S_u^{(i)} - 1 \right).$$

The event estimates are the sample proportions of $R_T^{(i)} < 0$ and $D^{(i)} < -0.20$. A mean has estimated MC error $s_R/\sqrt{M}$; an event probability has estimated error $\sqrt{\hat{p}(1-\hat{p})/M}$. Errors shown as percentages are on the same scale as their estimates (percentage points for probabilities).

Table 3: Conditional two-year Monte Carlo outcomes.

Metric	RW	GBM	Log-OU
Mean terminal change	178.73%	68.22%	36.69%
Median terminal change	0.75%	55.33%	35.66%
5th percentile change	-90.17%	-19.32%	10.43%
95th percentile change	939.40%	199.56%	66.62%
Probability of terminal decrease	49.71%	13.68%	0.70%
Probability of drawdown exceeding 20%	100.00%	81.23%	93.26%
MC standard error: mean	7.33%	0.70%	0.17%
MC standard error: decrease probability	0.50%	0.34%	0.08%
MC standard error: drawdown probability	0.00%	0.39%	0.25%
Analytical mean terminal change	171.83%	67.78%	36.25%

The 5th–95th percentile bands are pointwise simulation intervals, not simultaneous path coverage or confidence intervals for market outcomes. The convergence check uses nested samples of 200, 1,000, 5,000 and 10,000 paths. Error bars are approximate ±1.96 MC standard errors for the mean and terminal-decrease probability, not measures of model uncertainty. No error bars are assigned to the 5th percentile in this comparison. Analytical means benchmark the implemented updates, including Euler OU.

The same Gaussian shocks are reused across models for comparisons; paths within each model remain independent. Increasing the path count reduces simulation error but cannot remedy model misspecification or parameter uncertainty. Normal error approximations can be unreliable near probabilities zero or one; zero sampled events do not establish impossibility. The additional paths do not enter the ADF tests.

AAPL: Monte Carlo figures

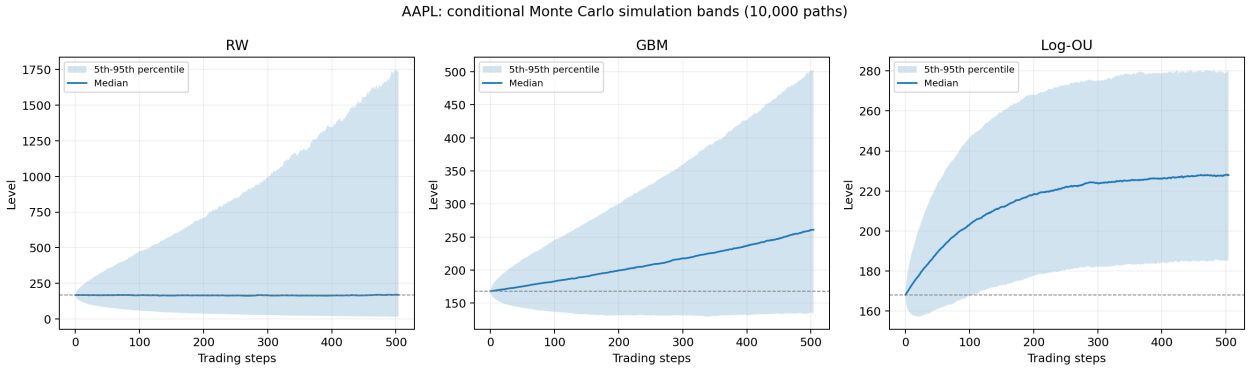


Figure 5: AAPL: Median levels and pointwise 5th–95th percentile bands.

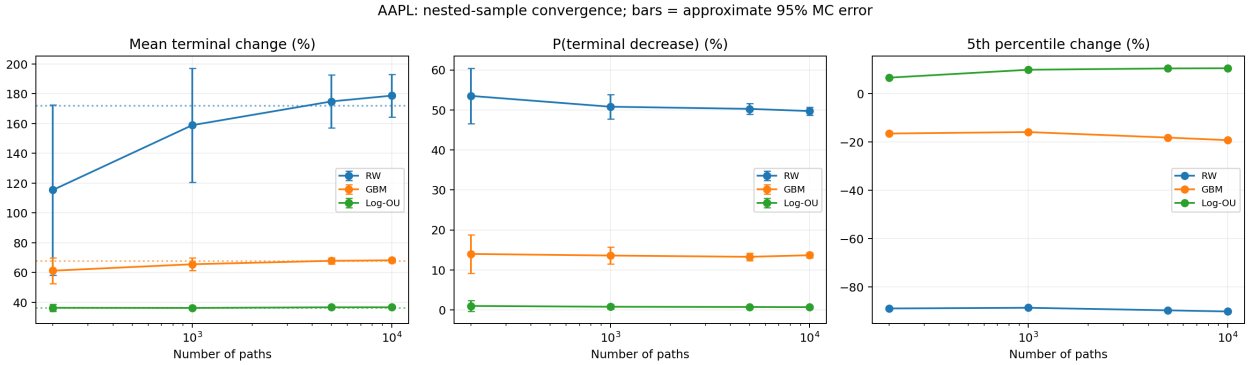


Figure 6: AAPL: Nested-sample convergence. Dotted lines in the mean panel are analytical expectations; error bars show simulation error only.

4 Results: VIX

Table 4: VIX: observed and simulated statistics.

Metric	VIX	RW	GBM	Log-OU
Annualized mean log return	9.10%	0.47%	20.03%	4.68%
Annualized volatility	141.47%	99.74%	141.77%	141.87%
Return autocorrelation (lag 1)	-0.05712	-0.0003034	-0.00425	-0.006696
ADF p-value: returns	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$
ADF p-value: price series	0.0002519	0.4088	0.3868	0.2397
Return skewness	0.9248	-0.001577	-0.005083	0.004244
Return kurtosis (Pearson)	10.11	2.994	2.973	2.997
Maximum drawdown	-74.26%	-77.41%	-84.70%	-83.60%

The sample contains 504 observed daily log returns. Its annualized mean log return is 9.10% and annualized volatility is 141.47%.

4.1 Distribution and risk

Observed Pearson kurtosis is 10.11, versus 2.97–3.00 across the simulated models. Observed skewness is 0.925. Gaussian shocks do not reproduce the observed higher return moments in this run. Observed maximum drawdown is -74.26%; model drawdowns are path averages and should not be treated as forecasts.

4.2 Persistence and model fit

Lag-one return autocorrelation is -0.057. This is a descriptive measure, not a formal test of market efficiency. The ADF test on observed log prices rejects the unit-root null at the 5% level. That result alone does not identify an OU process. GBM simulated volatility is 141.77%, while its annualized mean log return is 20.03%. The log drift is calibrated to the sample mean; finite simulation samples still introduce Monte Carlo error.

RW: geometric random walk; GBM: geometric Brownian motion; Log-OU: exponentiated OU process.

VIX: Price dynamics

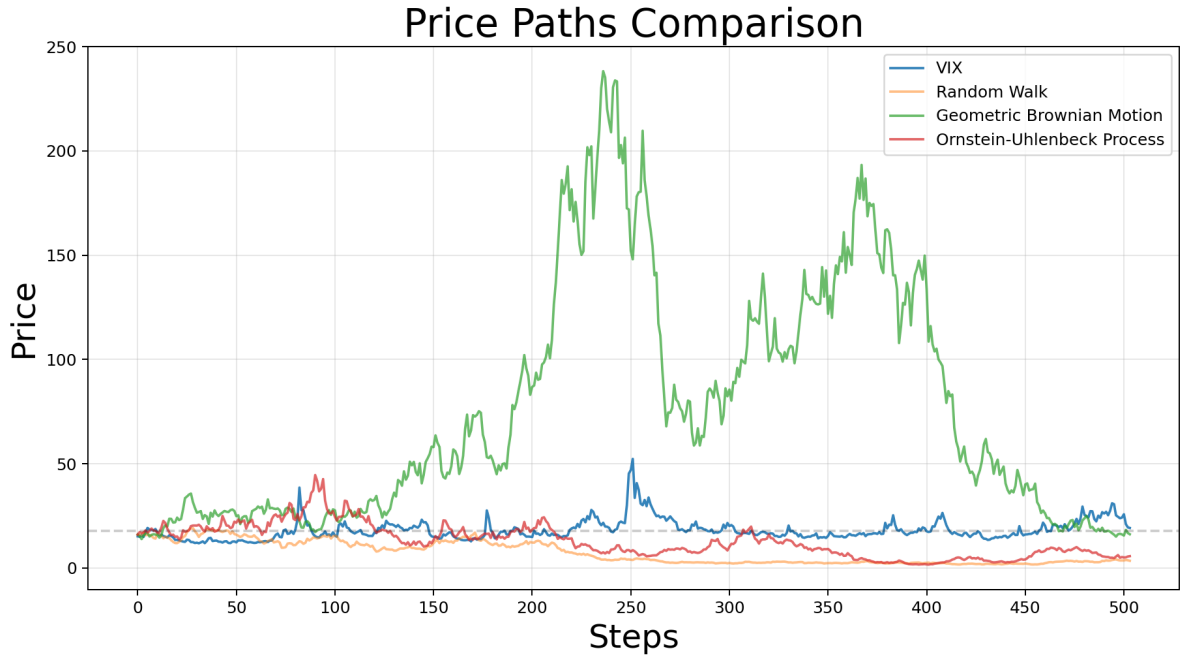


Figure 7: VIX: Observed series and one illustrative path per model.

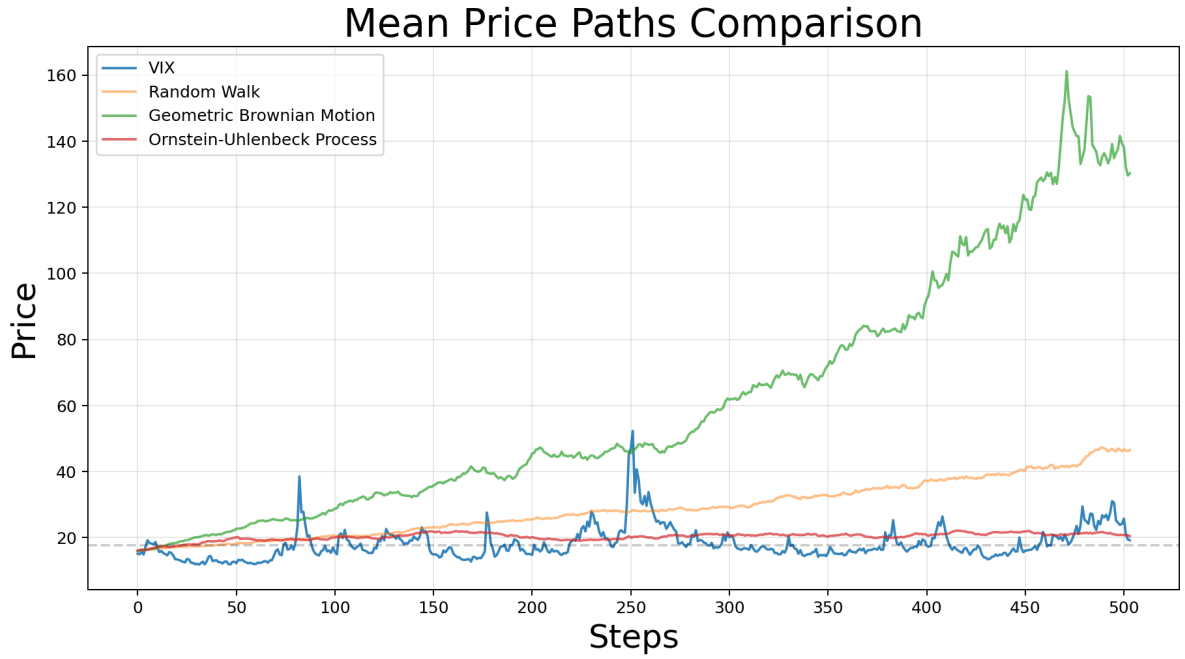


Figure 8: VIX: Observed series and ensemble mean of 200 simulated paths.

VIX: Return diagnostics

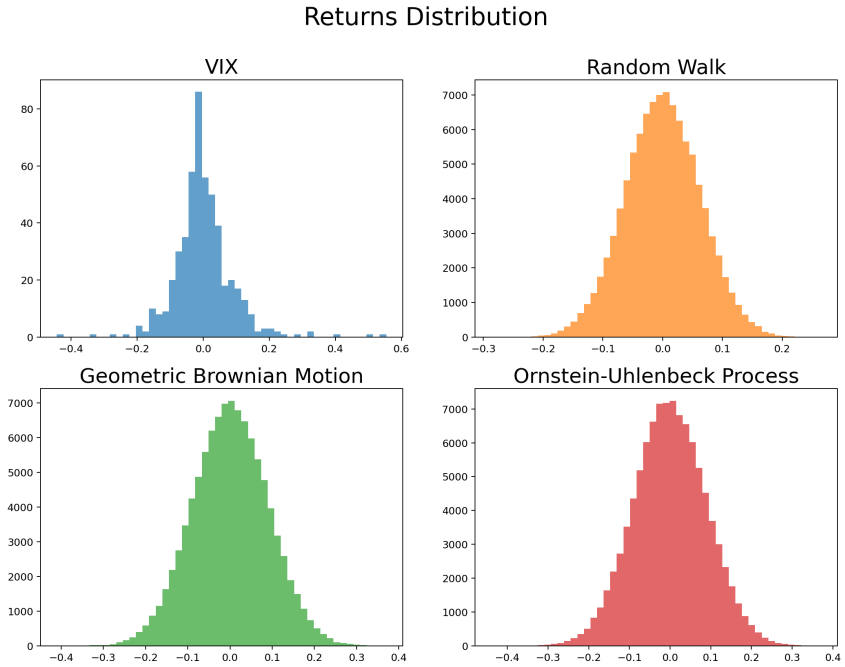


Figure 9: VIX: Observed and simulated log-return distributions; counts use different sample sizes.

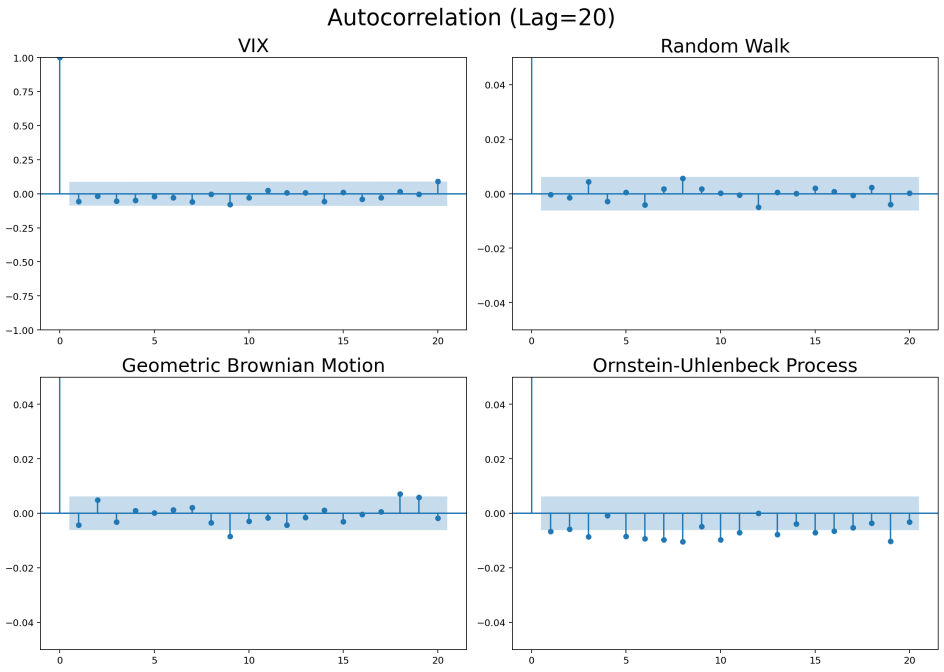


Figure 10: VIX: Log-return autocorrelations; simulated diagnostics pool independent paths.

VIX: Monte Carlo uncertainty and risk

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The event estimates are the sample proportions of $R_T^{(i)} < 0$ and $D^{(i)} < -0.20$. A mean has estimated MC error $s_R/\sqrt{M}$; an event probability has estimated error $\sqrt{\hat{p}(1-\hat{p})/M}$. Errors shown as percentages are on the same scale as their estimates (percentage points for probabilities).

Table 5: Conditional two-year Monte Carlo outcomes.

Metric	RW	GBM	Log-OU
Mean terminal change	178.73%	841.52%	32.34%
Median terminal change	0.75%	21.24%	11.74%
5th percentile change	-90.17%	-95.50%	-56.27%
95th percentile change	939.40%	3192.47%	189.44%
Probability of terminal decrease	49.71%	45.67%	41.76%
Probability of drawdown exceeding 20%	100.00%	100.00%	100.00%
MC standard error: mean	7.33%	63.25%	0.85%
MC standard error: decrease probability	0.50%	0.50%	0.49%
MC standard error: drawdown probability	0.00%	0.00%	0.00%
Analytical mean terminal change	171.83%	787.72%	30.39%

The 5th–95th percentile bands are pointwise simulation intervals, not simultaneous path coverage or confidence intervals for market outcomes. The convergence check uses nested samples of 200, 1,000, 5,000 and 10,000 paths. Error bars are approximate ±1.96 MC standard errors for the mean and terminal-decrease probability, not measures of model uncertainty. No error bars are assigned to the 5th percentile in this comparison. Analytical means benchmark the implemented updates, including Euler OU.

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VIX: Monte Carlo figures

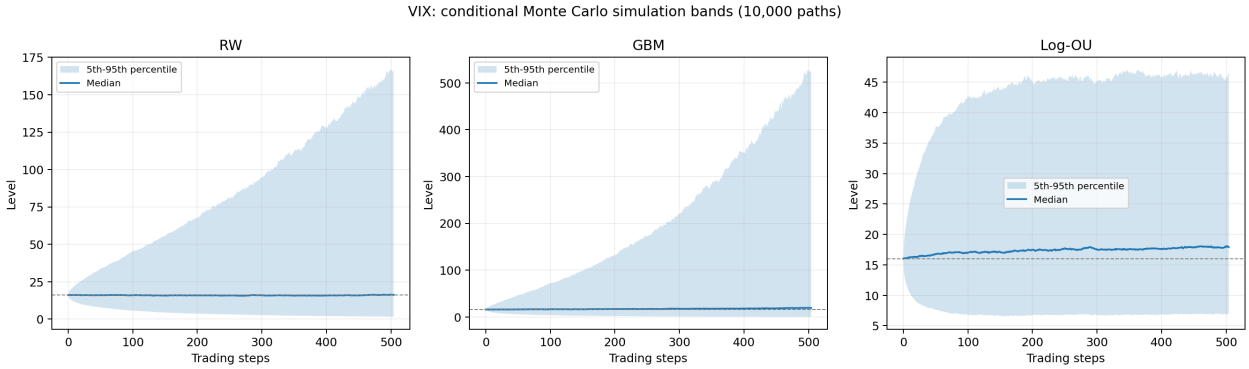


Figure 11: VIX: Median levels and pointwise 5th–95th percentile bands.

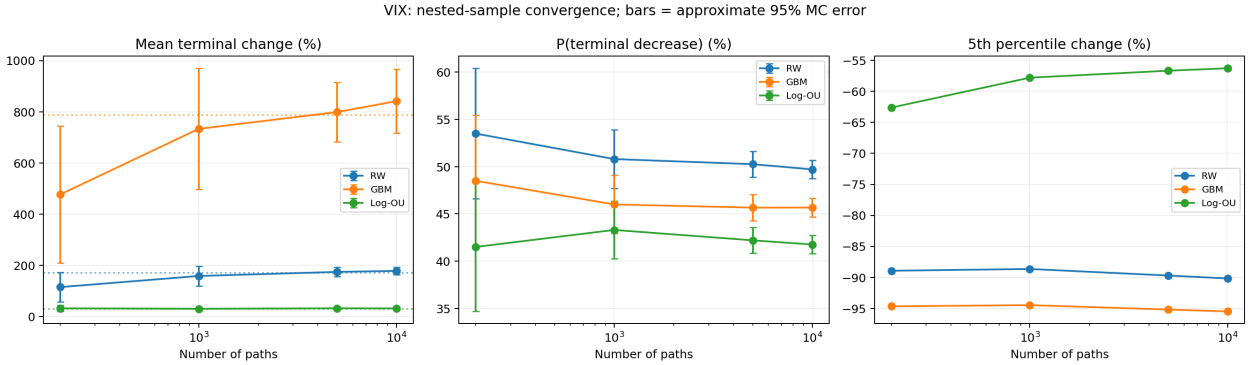


Figure 12: VIX: Nested-sample convergence. Dotted lines in the mean panel are analytical expectations; error bars show simulation error only.

5 Discussion and conclusions

Table 6: Observed diagnostics across the two assets.

Asset	Return ACF(1)	Pearson kurtosis	Log-price ADF p-value
AAPL	0.052	13.77	0.1798
VIX	-0.057	10.11	0.0002519

H1: Limited evidence on dependence. The observed autocorrelations and ACF plots describe serial dependence, but there is no formal joint whiteness or market-efficiency test in this implementation. The broader efficiency hypothesis remains unresolved.

H2: Limited random-walk fit. The implemented random walk uses unit annual log volatility rather than an asset-calibrated scale. Its risk and distributional fit must therefore be assessed relative to this specific specification, not as a rejection of every random-walk model.

H3: Distributional mismatch. Both observed series have higher kurtosis than the approximately Gaussian simulated returns in this run. GBM reproduces its volatility input closely but does not reproduce these tail characteristics. Its log drift is calibrated to the observed mean, with remaining sampling differences measured by Monte Carlo error.

H4: Stationarity is not model identification. The asset-level ADF results give evidence concerning a unit root under the test's default specification. Neither rejection nor non-rejection alone establishes that an OU process is suitable; θ is not estimated here.

5.1 Scope and further work

No single implemented model captures all observed features. The main lesson is to compare scale, tails, and persistence jointly, rather than judge models from one illustrative path or a matching volatility. These results are descriptive, not investment or forecasting evidence.

A stronger follow-up would preserve dated observations, align simulated and observed price grids, use consistent price transformations for ADF, compute dependence diagnostics per path, estimate mean-reversion speed, and evaluate the calibrated models on an unseen sample. These are proposed extensions, not changes made for this report. The new Monte Carlo experiment quantifies conditional outcomes and numerical precision, rather than testing out-of-sample predictive accuracy.